

AUTODOCSv2

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AUTODOCS: A VOICE-POWERED DOCUMENT GENERATOR FOR SEAMLESS TRANSCRIPTION AND AUTOMATION

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CHAPTER I

INTRODUCTION

The need for documentation that is accurate, timely, and efficient is extremely important in our day and age of quick communication and fast digital collaboration. The traditional methods of note-taking and manual typing lack speed and coherence and reliability for multilingual environments and rapid transformation environments according to the study by Xu et al. in 2024. Artificial intelligence technologies that develop at a quick pace now provide new opportunities for organizations to automate their meeting and presentation documentation which includes automatic speech recognition and natural language processing systems according to Georgiou in 2025. The new tools create real-time transcription together with language support and deep contextual understanding while they document content with improved reliability and smoother functions. The new innovations eliminate long-standing inefficient processes which support better organizational productivity and decision-making according to the study by Jelassi et al. in 2024.

The absence of effective documentation systems is, all the more, an impediment to proper communication globally and locally. Many organizations deal with inconsistencies in manual record keeping and language barriers along with time-consuming processes for documentation (Alharbi et al., 2022). In case of multilingual countries such as the Philippines, these are further complicated by frequent code-switching, inadequate supply of trained manpower, and weaknesses in digital infrastructure (Muntendam & Parafita Couto, 2024). A large number of institutions are still relying on handwritten notes, which are subject to tremendous and easy loss and distortion. These kinds of issues slow down the flow of information and hinder

organizational memory and major workflows. As a result, communication gets interrupted, and decisions lose their ability to be made efficiently.

Instances in which documentation problems are evident are many in the Philippines, such as in public offices, schools, and small-to-medium enterprises, where handwritten or manually typed reports are often found (Briones et al., 2023). Compounding the problem of documentation are insufficient internet connectivity and very limited professional transcribing tools, among others (Elhadad et al., 2024). Meetings that switch back and forth between English, Filipino, and the regional dialect are more suited to certain mannerisms. Incomplete records and lack of important details result from this scenario very often, thereby compromising the accuracy of documentation for absorbers of information (Rentillo et al., 2024). Long-term, these will have an adverse bearing on organization performance, transparency, and overall productivity.

The new systems use speech recognition technology together with AI potential language modeling technology to provide customers better contextual speech recognition solutions that can expand their operations. The speech recognition tools that provide multi-language support and offline capabilities become essential in areas where internet access remains unreliable. The different settings prefer these systems because they can identify speakers and understand mixed-language conversations and produce accurate transcripts (Hai et al. 2025). AI-powered language models use their intervention to create conversation summaries and key point summaries and produce user-friendly documentation that improves raw transcripts into formatted documents. The two types of technologies combined further the documentation process to be faster, more accurate, and more inclusive, adapting to different linguistic and technological constraints.

Auto Docs is a voice-activated document creator that becomes the answer to such problems, using automated transcription and automation technology. The whole system of the intelligent documentation assistant involves tying next-in-class speech recognition with state-of-the-art language modeling. Because of this feature, Auto Docs will have an easy, converging access into various end uses, like education, government, and business. Users will have the ability to upload audio files with specific numbers of speakers attached, create transcripts, identify speakers, summarize, and export documents in multiple formats. With this exceptional platform, institutions will be able to make their documentation faster, smarter, and more reliable. The future changes in how such information is collected, stored, and made available for use have begun.

Project Context

The efficient, accurate, and effective documentation of meetings and presentations is an essential part of communicating, transparency and efficiency in organizations. Much documentation processes in most sectors in the Philippines such as education, government, small-to-medium enterprises are still manual and would probably expect delays, human error, and mostly, loss of information (Bustillo & Aguilos, 2022). Most importantly, environments with limited manpower or closely packed schedules need precise recording of discussions, decisions, and action items. Poor documentation can cause loss of account, hamper the continuity of workflows, and lessen the impact of institutional memory, ultimately affecting overall performance (Isbasoiu & Volosevici, 2025).

The challenge is often encountered in academic and business meetings, student assemblies, research defenses, and organizational planning sessions. Nevertheless, appointed recorders usually get lost in lengthy discussions, quick exchanges, or multi-speaker dialogues (Wagner et al., 2025). Most transcriptionists are either not trained in

the specialized skill or engaged in doing something else parallelly, which adversely affects their quality and timely completion (Gentle et al., 2024). The lack of any centralized or automated means of conducting live transcription or intelligent summarization also compounds the challenge, resulting in notes that are often not complete or are inaccurate, hampering decision-making and coordination on other projects (Lee et al., 2025).

It has been recognized by the Philippine higher education sector that to improve communication and enhance productivity, digital tools and AI must be strangled together (Alenezi et al., 2023). However, financial barriers and the lack of local language support have kept the evolution towards fully digitalized processes quite slow, along with the technical complexity in the existing products (Atobishi et al., 2024). The advances of speech-to-text technologies are still not much useful in the Philippine setting because most of them do not address the Filipino language context, and some do not work without a continuous internet connection or expensive subscriptions. This underlined the emerging requirement for documentation systems that are accessible, inexpensive, offline capable, and inclusive of languages (Espartinez, 2025).

To solve this problem, any of the modern information and communication technologies like OpenAI Whisper offer good solutions. Whisper is an offline speech recognition toolkit that supports dozens of languages, including English and Filipino, and is quite capable of generating text from spoken dialogue (Karibayeva et al., 2025). Since it can work without a stable internet connection and is compatible with low-resource devices, it is ideal for Philippine institutions. Because it can automate transcriptions of live discussions or those from recorded audio, it greatly lessens the burden of manual work and improves reliability and efficiency in documentation (Cabello-Collado et al., 2024).

Although large language models (LLMs), like the GPT-based platform provide sophisticated natural language processing functions, including complementary transcript summaries, extractions of ideas, speaker identification, and well-formulated writes from raw unstructured audio input, speech recognition still predominates (Gómez-Sirvent et al., 2025). The combination of LLMs with OpenAI Whisper offers a better voice input-to-polished written output documentation pipeline for organizations and revolutionizes how they capture, interpret, and share meeting content. This combination improves report accuracy and consistency and enhanced accessibility to reports across sectors (Alawida et al., 2023).

Autodocs is intended as a one-stop shop that integrates OpenAI Whisper and LLM capabilities into a clean and easy-to-use platform for the automating of meeting and presentation documentation. Designed with the Philippine educational and local organizational needs in mind, Autodocs allows users to upload audio recordings, transcribe speech identifying speakers, summarize the discussion, and export documents in various formats. The process of documentation from start to finish becomes automated through autodocs which enables users to concentrate on their collaborative and strategic work while the system maintains exact information capture and organized storage and easy retrieval of vital data.

Research Objectives

General Objective

This study focuses on developing “AutoDocs: A Voice-Powered Document Generator for Seamless Transcription and Automation”

Specific Objectives

1. To develop a system that enables users to:

- 1.1. Live audio recording
- 1.2. Import existing audio files
2. To integrate speech-to-text transcription with number of speaker identification.
3. To implement features that allow users to:
 - 3.1. Edit transcripts for accuracy and clarity.
 - 3.2. Summarize content into concise reports.
 - 3.3. Translate documents supporting both English and Filipino languages.
4. To provide functionalities for exporting reports to various file formats and tracking meeting duration.
5. To evaluate the developed system using the Technology Acceptance Model (TAM), focusing on perceived ease of use, perceived usefulness, attitudes toward use, and behavioral intention to use.

Conceptual Framework

According to Figure 1, the AutoDocs Conceptual Framework consists of a complete, modular system for converting raw audio data into a speaker-aware, linguistically-adaptive summarized transcript. The framework comprises a series of processes ranging from audio acquisition to advanced natural language summarization, integrating traditional signal processing and state-of-the-art machine learning models. This automation will ensure that documentation workflows will be handled in sharpest relief and clarity, in addition to supporting multiple languages, making it ideal for training, reporting, and instruction applications.

Inception of the process starts with a user giving an audio input, which may be from a live recording or from a file that is uploaded. This input is treated by a front-end module in charge of importing audio, starting recording sessions, and the acceptance of different file types. The audio recording process sends complete audio data to main

storage so the system can track its movement and users can access it for future work. The storage layer functions as a temporary storage solution which enables the entire system to run different operations at the same time while it supports multiple users and different session requirements.

The audio materials are stored and then subjected to further preprocessing to extract some useful content. The process includes dedication which helps find important parts of audio content, audio separation which enables the user to remove background sounds and overlapping audio, and speaker separation which allows the user to identify different speakers in the audio content. The stage produces a dataset which contains cleaned audio segments that the Dataset Audio module will receive. The module creates speaker-specific data streams for transcription to ensure that the transcription engine receives input which is both clean and properly organized.

The transcription process relies on the Whisper Model which functions as a neural encoder-decoder system that transforms audio signals into written text while recognizing the spoken language. The model performs deep audio feature extraction and generates a raw transcript that retains speaker identity information. The alignment process matches each text segment with its associated speaker to create a transcript which includes speaker identifiers and time coding. The report enters two distinct processing paths toward its two destination modules which include a Summarization system and a Large Language Model (LLM). The two functions serve divergent purposes but they can both utilize the same data for their respective operations.

The Summarization module enables speaker differentiation through precise linguistic and structural transcript enhancements which include Filipino-English translation and final transcript formatting for fundamental summary tasks. The LLM module uses advanced methods to process data which includes methods of tokenization

and context windowing and prompt engineering. The content of the annotated transcript will undergo a summarization process which uses the contextual information to produce summaries. The finished document presents a refined multilingual text which respects speaker identity and can be used as training materials or meeting notes or educational resources. AutoDocs demonstrates a modular intelligent document system which provides audio-to-actionable text conversion capabilities through its scalable design and advanced machine learning technology.

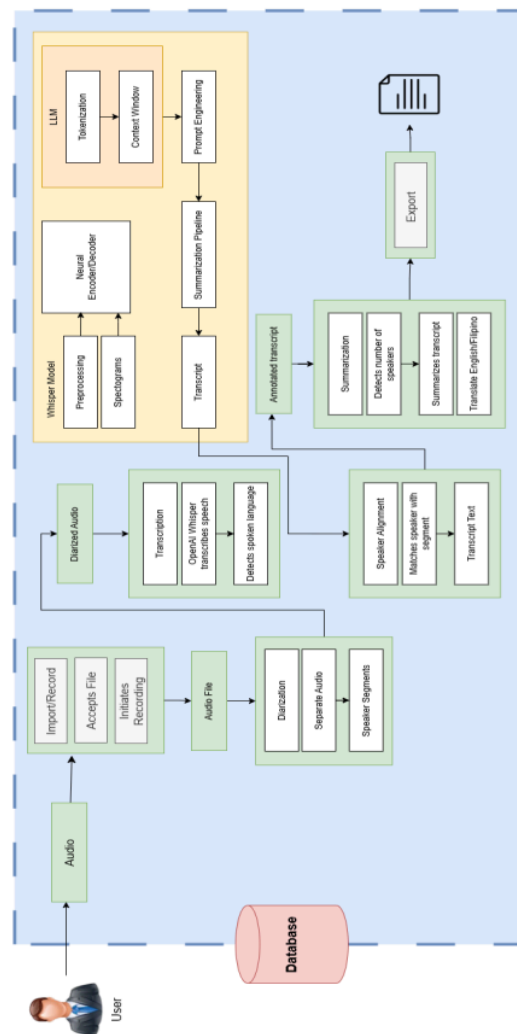


Figure 1. AutoDocs Conceptual Framework

Project Purpose

The study benefits from a diverse group of people who work with meeting documentation processes in both educational and professional settings. The system would provide a dependable transcription and summarization solution which enables users to transcribe and create summaries while overcoming obstacles from manual note-taking and reducing mistakes in official documents. Some benefits include improved productivity, guaranteed consistency in documentation, and support in maintaining accurate records to aid in decision-making and compliance for all secretaries, administration staff, teachers, school administrators, student organization bodies, IT professionals, and the prospective developers. It affords a customized solution addressing specific documentation needs for each target group while also serving as a reference for more research and innovation in the AI-assisted documentation system.

The beneficiaries of this study include:

Secretaries and Administrative Personnel. Enhancing an efficient tool that would automate the transcription and summarize spoken content during meetings, thus relieving manual burden and improving the accuracy of official records.

Teachers and School Administrators. Assisting in keeping the records for consistent and accountable faculty meetings, seminars, and training sessions that would lead to proper functioning of the school's planning, reporting, and compliance tracking.

Student Organizations and School Councils. To aid student leaders in the generation of the complete and clearest meeting documentation in good time for submission or project reporting to the school authority.

IT Professionals and Documentation Officers. The purpose of this system is to create a one-stop shop for managing transcripts, reviewing their summary content,

and exporting their documents for auditing or compliance purposes, as well as institutional archiving.

Future Developers and Researchers. Be a model for implementing open-source speech-to-text and LLM technologies in real-world productivity tools, and further innovation in AI-assisted documentation systems.

Scope and Limitation of the Study

AutoDocs is a mobile application that automates all forms of documentation for meetings or presentations, including but not limited to schools, government offices, and businesses. This system is geared toward secretaries, documentation officers, and, of course, administrative staff in the office, which will help them accurately record and present their reports of meetings. AutoDocs is equipped with advanced ICT tools including OpenAI Whisper for transcription and speaker identification of audio signals, with support from large language models (LLMs) as the ultimate tool that will summarize the transcriptions and finally present a well-structured, downloadable report. It also features a Bi-lingual Application for Support on English and Filipino language, audio recording, audio file import, edit transcript, content translation, meeting duration, and speaker detection functions. Optimized for cross-platform mobile deployment, Access to AutoDocs is easy and efficient within a lively fast-paced environment.

AutoDocs has a number of limitations, despite its fine functionality. For instance, speech recognition and speaker identification accuracy are greatly influenced by audio clarity, and background noise, overlapping speech, or unclear recordings would have to greatly affect transcription outcomes. While the system supports both English and Filipino, performance could wane because of regional accents, assimilation caused by code-switching, or the informal language that we commonly use when

talking. Speaker diarization or distinguishing between more than one participant might not be too reliable in large or complex meetings. The LLMs-based summarization feature requires an active internet connection; hence, it would not be fully functional in offline or poorly connected areas, together with all its other features. The advanced features which include contextualized translations and highly specialized domain-specific summaries need fine-tuning so they can achieve better consistency.

REVIEW OF RELATED LITERATURE**Review of Related Studies***Manual Documentation*

Documentation functions as a vital resource which enables organizations to maintain effective communication and responsibility tracking and operational efficiency (Persson et al., 2021). The accurate documentation process will improve decision preservation and action item tracking while establishing common understanding of expected results (Ademola et al., 2024). The discipline enables all parties to understand essential aspects of reality which exist beyond their current knowledge. Meetings, as the principal avenue for collaboration, will ultimately require decent documentation to grant continuity and visibility in informing decision making in any given organization (Da Silva & Boton, 2025).

Due to the significant challenges posed by manual documentation, it has often been assumed that certain key points may be lost by note-takers on distractions, fatigue, or multitasking, while along with that, differences in writing styles may further contribute to inconsistent or unclear records (Dunn, 2021). It is painstaking converting the spoken word into a written document, thus extending the workflows of the organization and, in turn, the completion of the project (Khahro et al., 2023). In high-stakes situations-medicine, education, or government, for instance-these inefficiencies lead to misinterpretation, loss of information, and diminished accountability (Kovari, 2024).

Moreover, the manual documentation was not addressed by linguistic diversity and accessibility needs. The combination of speech barriers and accents and fast speech creates transcription difficulties which prevent hearing impaired people and those with limited language skills from understanding spoken content (Stitzinger, 2025). People who take

notes according to traditional methods do so without a standardized system because their personal approach to note-taking differs from their choice to record content at exceptional standards. The existing deficiencies demonstrate the requirement for documentation tools which deliver precise and constant documentation across all user groups (Jenko et al., 2025).

Automated transcription and AI-based summarization serve as technological solutions which businesses require. The current utilities enable users to convert spoken language into written text, which they can use for more than 45 different languages while the system identifies who is speaking and creates organized summaries of the dialogue (Di Leo et al., 2025). The combination of speech recognition systems and advanced language models enables users to understand context while disambiguating meanings and producing standardized documentation which is both accurate and user-friendly (Tejedor-García et al., 2021). The technologies help organizations to achieve higher productivity because they reduce manual tasks while creating dependable records which support critical decision-making processes (Murire, 2024).

The study establishes direct links to practical applications of automated solutions through its research findings. The research focuses on developing an AI-based documentation platform which will solve all the issues that arise when people create manual records through their repeated errors and their time delays and problems with document access. The research team will use speech recognition and language processing technologies to build a system which produces meeting records that are both contextually accurate and user-friendly. The system streamlines documentation work while it protects essential data through organized storage which enables quick access to information needed for later use.

Quality of Meetings

The organization needs to understand how meeting quality affects team work efficiency and communication and decision processes. Structured meetings help participants to take part in the meeting while they establish clear communication and create a space for participants to work effectively (Paredes-Saavedra et al., 2024). Unstructured and lengthy meetings result in people losing interest and becoming confused which results in wasted time. The success of our meetings depends on three essential elements which include a clear agenda and time management and maintaining concentration during meetings (Valente et al., 2024). The combination of proper facilitation together with correct note-taking results in clear decision-making and action item identification which establishes complete responsibility for all parties (Aldemir & Uçma Uysal, 2025).

Most organizations keep facing challenges to run effective meetings. The unstructured nature of those discussions results in disorganized speeches that lack both direction and coherence. Participants who attend the meeting may lose track of what they should do next when the meeting fails to deal with essential subjects (Fraboni et al., 2023). Manual documentation does not ease this situation because incomplete records together with inconsistent records create problems for subsequent actions while damaging institutional memory. The existing gaps lead to operational inefficiencies which result in major disruptions to the entire process of team collaboration (Pantiris et al., 2025).

The quality of meetings gets impacted by the process of post-meeting follow-up activities. Organizations fail to maintain their progress because they lack proper channels to share their decisions and tasks together with their outcomes (Navickiene et al., 2021). The documentation process leads to several problems because it results in people misinterpreting essential details which creates misunderstanding between parties.

Organizations need to create accessible meeting outcomes which provide precise information and actionable items to maintain their operational efficiency while sustaining project work (Lee et al., 2025).

Digital transcription tools together with AI-empowered assistants and real-time summarization systems function as practical solutions that improve meeting quality while ensuring precise documentation (Haz et al., 2024). Speech recognition and AI derivatives provide the opportunity for the delegates to be free from the spell of notes, but also pursue full engagement and contribution by eliminating the distraction caused, on one side, by note-taking-burdened participants, and on the other, by the potential misinterpretation potential in minutes captured.

With the integration of automated transcription and summarization tools, meetings are thereby effective in maintaining records of proceedings and results clearly stated, yet providing access to important facts. Indeed, through the application of these technologies, meetings become more structured: the participants do not hold attention and take part in the discussion at all during the decision-making process. It would lead to improved communication, more accountability, and higher performance levels of teams.

Accuracy of Transcription

Zuluaga-Gomez et al. (2023) argue that accurate transcription is key to effective communication, forming dependable records, and influencing decisions in organizations. According to Zemlickienė and her team from the year 2022, proper meeting transcription enables people to visit earlier discussions while they can see all decisions and action items without any possibility of confusion. Transcripts that contain mistakes or missing information lead to people misunderstanding their work duties which results in performance mistakes. The organization needs transcription accuracy because it helps

protect organizational data and promotes effective business operations according to Goldshtein and his team at 2024.

The process of manual transcription¹ faces major problems which create different degrees of transcription¹ accuracy problems. The combination of different note-taking methods and human mistakes plus lost information leads to documents that show missing or incorrect details according to Azadi and García-Peñalvo in 2025. Manual notation becomes more difficult when background noise disturbs the process or multiple speakers talk at once or speakers deliver information at a high speed according to Skoe and Kraus in 2024.

Organizations need solutions that produce reliable transcripts which maintain structured format while creating minimal work for staff because these solutions work as their complete record management system according to Necula and his team at 2024.

The introduction of AI-based transcription tools through newer technological tools has improved transcription services to achieve higher accuracy and reliability than all previous versions. The systems operate using two advanced technologies which deep learning and natural language processing use to enable speaker identification and dialogue boundary detection plus resolution for difficult technical terminology according to Wu et al. in 2025. The system enables users to interpret their intentions while they research main topics and include specific words. The system creates transcripts which contain both precise information and additional important content. The tuning process uses conversational patterns with meeting contexts to delete errors while creating high-quality transcripts according to Punnaivanam and Velvizhy in 2024.

The transcription process gets disrupted by technical issues and environmental factors according to these two elements. Good-quality audio with perfect microphone placement and low background noise helps AI transcription systems achieve better results

according to Nogales and his team in 2024. Human review functions as a vital component which needs to handle sensitive and legal documents although automation handles most of the work according to Mohamed and his team in 2022. The process of post-processing transforms documents through four stages to create formatting changes which fix punctuation errors and add keyword tags and create summaries which enhance reading. The process of post-processing transforms documents through four stages to create formatting changes which fix punctuation errors and add keyword tags and create summaries which enhance reading. (Kaushal et al., 2024).

AI transcription tools provide organizations with verification functions which help them create correct documentation by combining contextual understanding with advanced document processing systems. The operation of these tools decreases manual work while they deliver total accurate records which help track decisions and progress through tasks. AI transcription systems present an efficient solution which improves organizational performance by providing trustworthy results that replace traditional manual note-taking methods.

Quality of Summarization

The main reason for effective summarization exists to develop essential concepts which maintain their original meaning so people can instantly recall meeting discussion points. Summaries let participants review central details of the meeting which helps them make choices and monitor progress on their assigned tasks without needing to read the complete transcript (Zhekova, 2025). A poorly generated summary could result in missing an essential detail which would lead to misunderstandings and misalignment of priorities and unmade decisions (Podobnikar, 2025). Thus, while good summarization may involve merely condensing the content, it should, however, retain the context, intent, and finer meaning, such that it becomes useful. With the summary being accurate, simple, and

actionable, the general communication within and outside the organization becomes beneficial and successful (Waseemullah et al., 2022).

Contextual accuracy is one challenge that many meet in summarization. If the summaries only compress the content, important topics, decisions, or implied meanings may easily escape their attention (Onan & Alhumyani, 2024). The type of meetings being attended, disorderly discussions, and failure to communicate well may lead to diminishing the quality of summarization even further (Subramaniam et al., 2022). Incongruity in the way that some aspects of the summaries will refer to an incapable recall for some participants on making choices or follow-up tasks which implies for some serious considerations into having summation processes that would be much robust and intelligent (Kuraś et al., 2024).

Artificial intelligence technologies from modern times provide solutions that improve summarization quality through their advanced capabilities. The AI systems can identify user intentions while recognizing essential subjects and specific industry terms to create summaries that match the given context (Miller et al., 2024). The tools connect to communication platforms to record all audio and visual elements together with additional contextual details which create a complete record of the conversation. The process of real-time summarization maintains participant interest by showing important details and delivering immediate responses which results in more efficient meetings and enhanced concentration (Gandrita, 2023).

The effectiveness of any summarization work depends on how well it implements personalization. Different participants need to choose which particular information they want to highlight, which includes action items and project milestones and high-level overviews (Peronikolis & Panagiotakis, 2024). The system enables teams to extract all relevant information through its bullet points and timestamps and contextual snippets which

create multiple paths to understanding. The features operate as a balance for executing brief actions because Alshami et al. (2023) explain this requirement.

Meeting transcripts become more accessible to users through AI summarizers which also enhance their accuracy and practical value. The systems process raw discussion material by understanding the context and processing it in real time while delivering user-based summaries which provide structured and actionable content. High-quality summarization, therefore, secures cooperation, decision-making, and performance in the overall organization-an indispensable component of modern documentation workflows.

Time Saved in Documentation

At the organization level, effective documentation is what facilitates productivity, accountability, and timely implementation of decisions. The manual process of recording meetings needs much time to complete yet results in multiple recording errors according to research by Khorsand and Peráček in 2023. With time, such delays are bound to reduce team efficiency and result in ineffective follow-up actions that slow down an entire project. Companies should ensure that discussions, decisions, and action items are recorded in a timely and accurate manner to make them agile and responsive (Moghaddasi et al., 2025). It is streamlined documentation that allows a team to focus more on collaboration or strategy tasks while decreasing time spent on administration (Ruiz-Rojas et al., 2024).

Manual transcription or even note-taking will consume a lot of time. A long discussion to be converted into readable structured documents can take hours, denying access to vital information (Rabbani et al., 2025). This time has most of the time forced on the staff who have to manage their documentation responsibilities with other work which causes much inefficiencies and heightened risk of errors. This has led to the indication that there is a requirement for solutions that shorten the time for the documentation process without making trade-offs on accuracy or completeness (Petitta & Ghezzi, 2025).

Automated transcription and summarization tools solve these challenges by significantly reducing into minutes the time needed to produce usable records for meetings. AI systems can transcribe verbatim, summarize discussions, and generate structured reports immediately upon conclusion of meetings. Tools that convert hours of spoken content into actionable notes in seconds free members from tasks of decision-making and participation to manual recording (Taş & Kiraz, 2023). Faster processing of post-meeting records allows decisions to be made and assignments given faster by the organizations (Zhang et al., 2025).

The documentation AI system enables users to save time while improving their ability to access content during meetings. Participants would stop taking notes during extended periods because they could maintain better focus for their conversations (Bollaert, 2025). Team members can begin content review upon obtaining accurate transcripts and summaries which provide immediate content access. The documentation process provides essential support for remote teams who need to work across multiple locations and time zones (Ferreira et al., 2021).

The document automation system provides better efficiency because it delivers automated documentation processes which operate at fast speed and maintain precise results while delivering documents to users. The system creates an excellent additional workflow which increases engagement and delivers structured records at the right time to help the organization function effectively. AI-based transcription and summarization will ensure that vital meeting notes are quickly generated, enabling teams to act on decisions faster and remain consistently productive.

Error Rate in Final Documentation

The documentation process needs to keep its error rate at the minimum possible because this practice guarantees complete and accurate and clear and trustworthy records of organizational documentation. Organizations face the risk of seeing their operations

disrupted due to errors that occur from using incorrect transcripts or notes which lead to misunderstandings and wrong queries and mistakes during follow-up work (Amiraslani & Dragovich, 2022). The documentation process carries greater significance for decisions which hold higher importance in medical and legal and business contexts (Simion Luduşanu et al., 2025). The results of this process will create outcomes which guarantee accountability and enhance both decision-making procedures and organizational operational success (Holubčík et al., 2024).

Human errors in justice systems lead to the manual documentation process and traditional transcription methods suffering from mistakes which happen due to workers lacking focus or making mistakes because of personal conflicts or inadequate supervision of their work. Most notes contain incomplete information which leads to incorrect reporting of the information and the recording of more complicated discussions (Georgiou, 2025). The process of documentation faces difficulties because background noise overlaps with multiple speakers who have different accents. Researchers have serious doubts about the existing conventional methods which need advanced solutions that can create complete documentation without any errors (Younan et al., 2025).

The development of transcription and natural language processing technologies enables organizations to lower their documentation error rates. The advanced systems protect against errors through their ability to identify speakers and contextualize information and recognize field-specific phrases (Saadany et al., 2025). The techniques of real-time verification and multi-speaker modeling and acoustic-linguistic error correction all serve to enhance the accuracy of transcription results. The different systems make sure that their outputs remain consistent and clear while achieving near error-free performance in extremely complex or noisy meeting spaces (Serrano et al., 2025).

Human validation together with post-processing functions as a necessary element which ensures that essential documents maintain their accurate status throughout technological advancements. The transcript verification process begins with verifying speaker identification before workers check for any remaining mistakes (Almarshady et al., 2023). The combination of AI automation and human oversight creates a hybrid solution which increases documentation efficiency while producing documents that meet high standards of reliability and usability (Machucho & Ortiz, 2025).

The integrity of organizational records strengthens through reduced error rates in final documents which leads to improved decision-making. Organizations create trustworthy documentation through their use of AI-enabled transcription systems which provide contextual understanding and error correction capabilities to generate accurate and actionable documents. The processes strengthen our trust in recorded data while reducing the chance of misunderstandings and making sure important discussions and decisions are recorded as they happened.

Training Data of LLMs

Another thing to consider while training a model is the trade-off between the dataset size and the scale of the model. Models which become too large compared to the data available may not perform very well, while models which find themselves trained on insufficient or poorly distributed data are less likely to generalize properly (Althnian et al., 2021). Then careful consideration on dataset size, model parameters, and architecture distinguishes between optimal learning from wasting computation resources. Ideal scaling strategies will help Large Language Models (LLMs) to be fully effective without unnecessary expenditure of resources (Han et al., 2024).

Real and synthetic data may also be combined such that the performance increases when combined with appropriate handling. High-quality synthetic data, coupled with

proper curation and balanced with real data, can improve instruction-following, reasoning, and contextual understanding (Goyal & Mahmoud, 2024). Poor-filtered or badly generated synthetic data may decrease the training results. The ratio of diverse real-world sources of task-specific synthetic examples will maintain the models robust and adaptable (Meiser & Zinnikus, 2024).

Different ways, with the possible cases in which LLM Models have well-critical and balanced training datasets for the respective domains, have been organized. Thoughtful data preparation enhances accuracy, coherence, and generalization across tasks. And therefore, when quality, relevance, and diversity have been prioritized in training data, performance will increase in LLM, with greater correctness and context accuracy for automated documentation and language-based applications.

Related System

Table 1. Comparison of Systems, which reveals the capabilities of different transcription and documentation systems and compares their strengths and weaknesses in specialized fields, provides a general perspective on which systems can be comprehensive for recording, summarizing, translating, and exporting meeting content, as well as provide AI-powered summaries and speaker detection.

System 1 had features of recording, importing, summarizing, exporting, editing transcripts, knowing duration and speaker detection. It did not possess translation, however, preventing its functionality in multilingual environments. Despite this, the system's ability to generate an AI summary means that principal content can be blended and structured for human consumption.

System 2 had certain fundamental features like history tracking, audio import, file export, editing transcript, and awareness of duration with speaker identification. It lacks, however, features like audio recording, summarizing, translating, and giving AI summaries.

These deficiencies indicate the ability of this system to handle post-recorded contents, but it's difficult for real-time documentation, and multilingual support is given by this system.

System 3 had its strengths in recording, importing, exporting, editing, keeping track of duration, and speaker detection, but what it didn't have were summarization, translation, and AI-generated summaries. Therefore, more suited for ordinary transcription tasks as opposed to completely automated documentation, but overall it's a solid pick for those who just need a simple audio conversion to text without editing the transcript too much.

The features of **System 4** include everything necessary to record, import, summarize, export, and edit, as well as to track an interval and speaker detection. The only thing that limits the system's flexibility is that it does not translate-otherwise, it can generate automated summaries and would serve well in structured documentation for mostly monolingual environments.

The specifications of **System 5** seem fairly strong, including those related to the import, summary, translation, export, edition, duration tracking, speaker detection, and AI summaries. Recording was omitted, cutting off the option of live meeting capture. Its worth is more in the translation and AI summary development, which is why it is perfect for multilingual content and post-meeting reporting.

This system is excellent for an organization that desires multilingual transcription, as well as automated summaries, which are even more effective when it comes to managing workflows. **System 6** had an excellent feature set that includes recording, importing, summarizing, translating, exporting, editing, duration tracking, and speaker detection.

The recording of transcription is in Systems 7, 8, and 9. **System 7** offered importing, summarization, translation, exporting, editing, duration tracking, and speaker detection. But recording and AI-generated summaries were absent. That way, it suited but was far

from fast in transcribing and automatically summarizing live events into documents; hence, in general, productivity-wise diminishes.

System 8 just had importing, exporting, editing, duration tracking, and speaker detection as a feature set. It lacked recording, summarization, translation, and AI summaries, making it the least feature-rich system among the ones compared. Such a limited feature set is good only for basic transcript handling and not for documentation automation in full.

System 9 included recording, importing, summarizing, exporting, editing, duration tracking, and speaker detection but did not provide for translation. The AI summary function in the system increases content accessibility, but it may pose problems for users looking for multilingual support. This makes it useful for structured transcription work in the mainly monolingual environment.

Including, but probably not limited to, recording, importing and summarizing, transcribing, exporting, editing, tracking duration, and perhaps even speaker detection. Its excellent AI summary feature makes it effective for condensing content. Great functionality coverage makes this so terrific for teams that want automated transcription with multilingual options embedded.

AutoDocs provides the most complete set of features, from history to recording, importing, summarizing or translating functions, exporting to editing, duration tracking, speaker detection, and AI-generated summaries. It thus provides a complete end-to-end documentation solution, including live transcription, multi-lingual processing, and automated summary generation.

Table 1. System Comparison

App / Website	History	Record Audio	Import Audio	Summarize	Translate	Export to File	Edit Transcript	Know Duration	Speaker Detection	AI Summary
Otter.ai	✓	✓	✓	✓	X	✓	✓	✓	✓	✓
Rev.com	✓	X	✓	X	X	✓	✓	✓	✓	X
Temi	✓	✓	✓	X	X	✓	✓	✓	✓	X
Sonix.ai	✓	✓	✓	✓	X	✓	✓	✓	✓	✓
Trint	✓	X	✓	✓	✓	✓	✓	✓	✓	✓
Happy Scribe	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Descript	✓	X	✓	✓	✓	✓	✓	✓	✓	X
Fireflies.ai	X	X	✓	X	X	✓	✓	✓	✓	X
Airgram	✓	✓	✓	✓	X	✓	✓	✓	✓	✓
Supernormal	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
AutoDocs	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Synthesis

Organizations depend on manual meeting documentation which presents major difficulties in their functioning. The correct recording of discussions is vital to communication, accountability, and organizational efficiency because it sustains decisions, tracks action items, and ensures alignment among participants (Persson et al, 2021; Ademola et al, 2024; Da Silva & Botton, 2025). People make mistakes during note-taking which creates improper records because they become distracted or tired or their writing patterns differ from one another (Dunn, 2021; Khahro et al., 2023). The process of documenting meetings faces obstacles from multilingual requirements and accessibility needs which create understanding difficulties that prevent complete meeting content capture (Stitzinger, 2025; Jenko et al., 2025). Organizations require tools which can

automatically handle transcription and summation tasks to create accurate meeting records because current documentation methods have too many deficiencies.

The remarkable success of AI transcription and summarization systems proves that technology-based solutions can successfully resolve existing problems in this field. OpenAI Whisper is one of these examples whereby speech recognition technologies translate spoken language into precise real-time text which relieves manual labor and provides support for all interacting languages and speakers (Mukhamadiyev et al., 2022; Lin et al., 2024; Rodriguez et al., 2025). The combination of Have's ¹³ Large Language Models (LLMs) with Large Language Models (LLMs) enables users to connect contextual information to key details which they can transform into structured summaries that follow their original meaning (Peykani et al., 2025; Zhang et al., 2025; Sternad Zabukovšek et al., 2022). The merging attributes of transcription, summarization and translation make sure that the meeting documentation is probed timely, standardized and accessible to informed decision making and collaboration within organizations (Vandeghinste et al., 2019; Emil & Remus, 2025).

Automated documentation also enhances the quality of meetings appreciated, ensuring accurate transcription and improved organizational efficiency, generally. With the complete record of discussions, participants can realize full engagement without the distractions of note-taking and minimization of error and improvement of follow-up actions (Zuluaga-Gomez et al., 2023; Haz et al., 2024; Bollaert, 2025). State of the art summarization tools condense discussion, but without losing context or critical detail, to quicken review and better decision making (Zhekova, 2025; Peronikolis & Panagiotakis, 2024). Further, integrated transcription aware in translation is endowed for multilingual environments and differences to be clear among all participants (Haque et al., 2019; Mohammed, 2025). The AI-supported solutions which work together as one system, create

a complete framework which helps organizations improve their documentation processes. The system helps organizations document their work better, while it reduces mistakes and delays, and it enables staff members to work together more effectively.

METHODOLOGY**Research Design**

The research used a descriptive research design to systematically define the AutoDocs application's features and functions and its perceived effectiveness. The descriptive research design establishes its foundation through the observation of actual application performance with documented operational results which include transcription and summarization and translation and reporting functions. The study documented how users interacted with the system by measuring their experience with the system and their ability to use it and their overall satisfaction with the system. The researchers used survey and structured evaluation forms to examine how administrative staff and secretaries and documentation officers from the target user group perceived the platform based on their experiences with system usability and platform usefulness and their intentions to use the platform. The investigation revealed that AutoDocs successfully documented meeting and presentation requirements for educational institutions and government agencies and business organizations.

The study employed a developmental research design which focused on creating and developing and deploying the AutoDocs application. The system integrated OpenAI Whisper as its speech-to-text solution together with large language models for summarization and multilingual support for English and Filipino languages. The research used iteration testing together with user interaction evaluation to create a system that was both accurate and reliable and user-friendly. The study assessed application performance through descriptive and developmental methods while creating an efficient automated documentation tool that is simple for users to operate.

Population of the Study

Population studies include secretaries from various offices of Laguna State Polytechnic University – Siniloan Campus. The university employs 16 secretaries who work across 16 distinct offices. The staff members have the duty to record all meetings and presentations and official department communications. The research team selected these individuals as primary study participants because their complete record functions directly in both study and administrative activities which made them suitable users for testing the AutoDocs application performance and user experience.

Sampling Design

The sampling process used both the availability of people who participated in documentation setup and their direct involvement in documentation setup as its basis for selecting participants. Ten secretaries were selected as respondents from the Laguna State Polytechnic University – Siniloan Campus who were available and agreed to participate in the study out of 16 secretaries in total. The researchers used this approach to collect important information from people who had the highest understanding of the documentation process and its associated problems. The selected participants also represented various offices within the campus because this approach would capture different administrative documentation practices and perceptions from the entire campus. The study aimed to evaluate AutoDocs application usability and effectiveness and practical use through observation with 10 secretaries who participated in the study.

Data Collection Instrument

The study required data collection for assessing AutoDocs usage and its effectiveness evaluation among secretaries from Laguna State Polytechnic University in Siniloan Campus. The application provides administrative support through its functions that include audio recording and transcription of speech and speaker identification and speech summarization and translation and document editing and exportation. The instruments were designed to capture both qualitative data and quantitative data which assessed system features and usability and perceived system value.

To gather in-depth qualitative data on the experiences of secretaries in using AutoDocs, semi-structured interviews were conducted with selected secretaries. The interviewees explained their documentation methods while discussing their difficulties with traditional processes and their evaluation of application features. The open-ended questions provided respondents with the opportunity to share their experiences about specific functions which included speaker identification and transcript editing and summarization. The researchers gained better understanding of user needs and system strengths and areas which required development.

The structured survey questionnaire was distributed to all participants to conduct a quantitative assessment of the Technology Acceptance Model (TAM) through its standardized instrument. The questionnaire required participants to evaluate four assessment constructs which included Perceived Ease of Use and Perceived Usefulness and Attitudes toward Use and Behavioral Intention toward Using the system. The items also focused on key features in AutoDocs, such as recording, importing audio, transcribing, summarizing, translating, editing transcripts, estimating time, and exporting files. The respondents used a Likert scale to assess items which

permitted researchers to evaluate their perceptions and satisfaction level and probability of using AutoDocs in their regular work activities.

Statistical Treatment

The statistical analysis method evaluated AutoDocs based on Technology Acceptance Model (TAM) elements which were used to evaluate the application. The study evaluated Perceived Ease of Use (PEOU), Perceived Usefulness (PU), Attitude Towards Use (ATU), and Behavioral Intention to Use (BIU) by secretaries of the Laguna State Polytechnic University - Siniloan Campus. The researchers used survey questionnaires to collect data which they used to measure the mean and frequency and percentage of each survey item. The method assessed how users felt about the application while measuring their document work and their likelihood of using the application. The statistical treatment process established a complete evaluation of AutoDocs usability and practical performance through systematic assessment.

Table 2. Statistical Treatment

ANALYSIS	STATISTICAL TOOL
LSPU Siniloan Campus Offices' Secretaries evaluated the application in terms of:	
- Perceived Ease of Use (PEOU)	Mean
- Perceived Usefulness (PU)	
- Attitude Toward Use (ATU)	
- Behavioral Intention to Use (BIU)	

The Scale and Descriptive Rating Table uses mean values which will be converted into qualitative ratings through the application of a Likert scale. Expert feedback will use the interpretation table to establish AutoDocs overall quality and acceptability for assessment purposes. The quality rating will reach a level which corresponds to "Strongly Agree" when average ratings fall between 4.21 and 5.00. The rating range from 1.00 to 1.80 will show "Strongly Disagree" status while indicating system improvements need to occur.

Table 3. Likert Scale Interpretation

SCALE	RANGE	DESCRIPTIVE RATING
5	4.21–5.00	Strongly Agree
4	3.41–4.20	Agree
3	2.61–3.40	Neutral
2	1.81–2.60	Disagree
1	1.00–1.80	Strongly Disagree

Project Design

The voice-enabled application AutoDocs, designed for automating the documentation of meetings and presentations with OpenAI Whisper and LLMs, was guided by the Agile Scrum Methodology, which provided evidence of its iterative and adaptive development process through its application of Technology Acceptance Model principles. The Technology Acceptance Model, or TAM, indicates that users decide to adopt new technology based on their evaluation of two main factors, which are Perceived Usefulness and Perceived Ease of Use. The TAM framework components match with the agile framework because both systems use an iterative and adaptive development process. The short, defined sprint periods enabled the development team to choose which tasks should be tackled first based on feedback and usability testing and system function enhancements.

All in all, the development team consisted of three members, and each of them had streamlined, precise roles to enable collaboration and efficiency. The developer was coding core functionalities, integrating OpenAI Whisper for speech-to-text transcription and applying LLMs to summarize transcriptions and generate structured reports. User interface/user experience designer created easy-to-navigate, accessible, and visually clear layout-composed user interface to support easy-to-use quality. The documenter, who also served as Scrum Master, was responsible for sprint planning, task management, and timeliness of the project. At the end of each sprint, users selected from among the many office secretaries at the LSPU Siniloan Campus tested out the fully functional prototypes. Improvements and refinements that moved the system

toward an "ideal" state were guided by the feedback. The approach maintained AutoDocs alignment with TAM theory through three user satisfaction measures and two user perception measures and one application usage intention measure.

Software Development Model

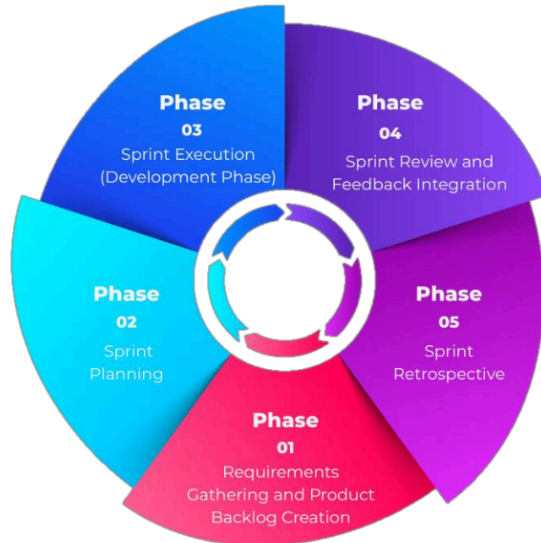


Figure 2. Agile Scrum Development Model

Requirements Gathering and Product Backlog Creation

The information collected was analyzed and made into a product backlog that closely follows the identified gaps and functionalities. Such key feature as transcription from speech to text, speaker identification, automated summarization, translation and report generation were prioritized in consideration with inputs from the secretaries. Every item in the backlog represented a user-centered need by reflecting the core problems and opportunities for improvement in the current documentation flow. Thus,

systematic documentation of these requirements will ensure that developers have a clear direction regarding design and implementation for AutoDocs that meets goals of the project to real practical needs of end-users.

Sprint Planning

The development team organized the work within the AutoDocs files during Sprint Planning by spelling out the tasks and features to be completed in each sprint. To plan, the team broke the entire system down into functional units, including but not limited to key real-time production quality speeches, speaker identification, summaries, and reporting. User feedback from the secretaries of LSPU Siniloan Campus, which was collected during the requirements phase for the project, formed the basis for what aspects were prioritized such that the system covered the real gaps in the existing process of creating minutes of meetings. The three members mutually apportioned the responsibilities where the developer would be encumbered with code and integration, the UI/UX designer with interface design, and the documenter with the coordination of tasks and documentation.

The two-week sprint was run like a tightly run ship with virtually very clear timelines for the iterative development process. At every turn of the iteration, things were very clear at the point where the team was going to set the objectives for that specific cycle, with examples focusing on development and integration of modules, testing functionalities, and enriching features based on user input. Daily stand-up meetings were organized for monitoring progress, failure issues as well as reviewing tasks where necessary. This continuous iteration of planning kept the team aligned with the project objectives and also led to timely delivery of the output.

At the end of each sprint, the team and select secretaries, who acted as representatives for end users, would review the progress with incorporation of feedback

into the next sprints. This allowed continuous improvement and alignment against user expectations on ease of use, usefulness, and reliability. The development team ensured with well-planned sprints and close collaboration that AutoDocs became a functional and effective tool for automating meeting documentation for LSPU Siniloan Campus.

Sprint Execution (Development Phase)

With the execution of the sprint at hand, the development team began to implement the features as defined during sprint planning under the agile scrum methodology. Much of the work to build, test, and improve the AutoDocs system was assigned on a collaborative basis by all members during daily stand-up meetings, which were used to critique progress, resolve problems, and adjust workflows and tasks to feedback and emerging needs. This was done to ensure that the development pathway was flexible, iterative in nature, and was addressed at all times to the stakeholders expectations, especially in response to the needs of secretaries at LSPU Siniloan Campus for accurate and efficient meeting documentation.

UI/UX design was created and refined through Canva, allowing for high-speed iterations over layouts of screens including audio upload, live transcription display, and summary panels.

React Native plays a frontend development framework in which the Canva designs are utilized to give birth to an active cross-platform mobile application. The frontend developer made it a priority to implement actions such as managing audio playback, viewing real-time transcripts, and navigating summaries. React Native offered smooth transition animations and an adaptive layout across both Android and iOS devices so that users can access AutoDocs and interact with its features seamlessly.

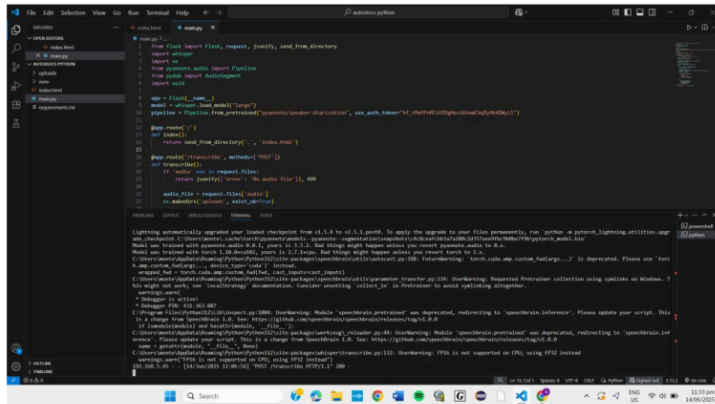


Figure 5. Backend Development

Sprint Review and Feedback Integration

The development team demonstrated AutoDocs functional capabilities through a Sprint Review presentation to selected users which included LSPU Siniloan Campus secretaries. The presentation focused on critical functionalities, for instance, audio-to-text transcription, live transcript display, and artificial intelligence-generated summaries of meetings. The system allowed users to test all its features which provided immediate results for usability testing and accessibility solutions and accuracy assessment. The team used this interaction opportunity to find out which system elements required enhancements so that the system would meet the real-world requirements of its target users.

The review team first collected all feedback through a structured process and then established priorities based on their potential user impact and system impact. The development team received documentation of all user interface suggestions which included both transcription accuracy and presentation of summaries. The phase established collaboration between end users and developers which provided the team

with standardized procedures for achieving better system performance. The secretaries shared their suggestions which revealed actual difficulties in recording meetings with multiple speakers and reducing the need for manual corrections which will undergo development in the upcoming iteration.

Sprint Retrospective

The development team conducted a retrospective session to examine their past sprint work by assessing everything that had occurred because their assessment included both positive and negative outcomes and potential future enhancements. The team members, which included front and back-end developers together with the UI/UX designer and Scrum Master, shared their thoughts about how the team should handle tasks and their workflow process and their team communication expectations. The retrospective process provided the team with a chance to recognize their achievements through successful OpenAI Whisper implementation and UI design work while identifying the areas which needed development.

The development process needed to understand three main challenges which were the delays that occurred because user feedback needed to be implemented, the problems which arose during multi-speaker recording sessions, and the three problems which occurred when AI systems produced summary content. The team developed strategies to resolve these problems by establishing new priorities for sprint planning and improving testing standards and adjusting model settings which would enhance transcription and summarization accuracy. The team built a culture which focused on development through their handling of obstacles when they worked to solve problems.

Testing and Evaluation Procedure

Testing activities followed the Technology Acceptance Model (TAM) framework to examine how end-users viewed AutoDocs technology. The primary purpose of this study assessed the two main factors which predict user adoption and satisfaction through Perceived Ease of Use (PEOU) and Perceived Usefulness (PU) assessment. The study involved LSPU Siniloan Campus secretaries who were asked to execute essential application functions through their work. The testing process used direct observation to monitor participant behavior while testing actual administrative duties through simulated scenarios which showed their daily work processes and challenges.

The study used monitoring tools to assess how participants operated their system. The study evaluated how users handled system functions and their system operation. The study used two methods to gather feedback from participants who used structured questionnaires and follow-up interviews. Participants gave their opinions about AutoDocs efficiency based on their experience with traditional manual documentation. The participants had all their essential system usage patterns evaluated through three main dimensions. The study used all collected data to assess system acceptance and potential adoption and needed system development areas. Researchers used response data to create a table which they used to identify four specific trends that showed people believed the system was useful and easy to use and they were satisfied with the system and they would keep using it.

Researchers used TAM evaluation to confirm that AutoDocs system functions operate correctly and its design meets user requirements. The evaluation process confirmed that all users could demonstrate their expected documentation efficiency

through the testing process which matched TAM standards. The study showed that OpenAI Whisper and LLM technologies can be used across multiple platforms to solve the documentation issues which affected previous administrative systems.

CHAPTER IV

RESULTS AND DISCUSSION

To develop a system that allows users to record live audio and presentations and import existing audio files

Live audio recording

The process of audio recording and transcription delivers a second major improvement which enhances efficiency and accountability and inclusivity. Team members can now participate in discussions without needing to write down essential points because the complete session will be recorded accurately (Suherman et al., 2025). The transcripts that follow become searchable and shareable, which enables users to access information instantly while they can use the material to recall important details about action items and decisions and significant events that occurred during the meeting. All this helps streamline follow-up, builds accountability, and ensures the preservation of deals and decisions for future reference, with clear links to those who made the commitments. Better collaboration, reducing mistakes, and improving the overall documentation flow will be achieved through organizations intending to combine audio recordings with automated transcription (Pardo et al., 2025).

As seen in Figure 6, the top section represents the home screen of AUTO DOCS, which presents users with a clear and intuitive interface to navigate the application. This home screen can be accessed by the audio recording interface via the "Add" button, thus bringing up the record audio interface for users to start capturing meeting or presentation content instantly.

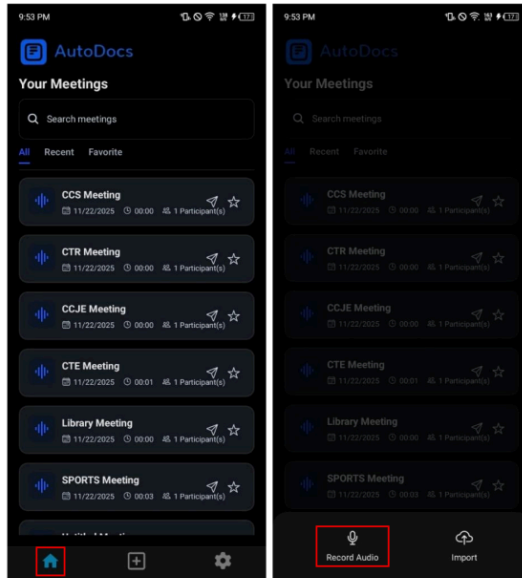


Figure 6. Homepage

After reaching the record-audio interface, the user starts the process by pressing the "Start Recording" button, capturing the spoken content live from that point. During the course of the recording, the interface allows for recordings to be paused or stopped, giving the user options in case they are interrupted or want to break the session into parts. With these options, the user maintains the full control of the recording process, allowing the precise recording of discussions without missing any important moment.

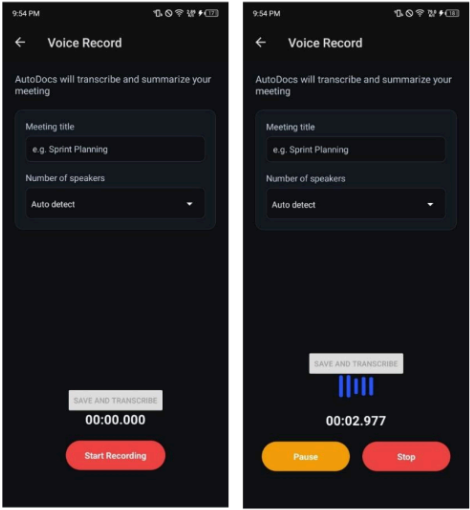


Figure 7. Recording

Users enter the needed data fields after performing the recording and, after that, they can save the recording along with its data, storing audio securely and later transcribing it.

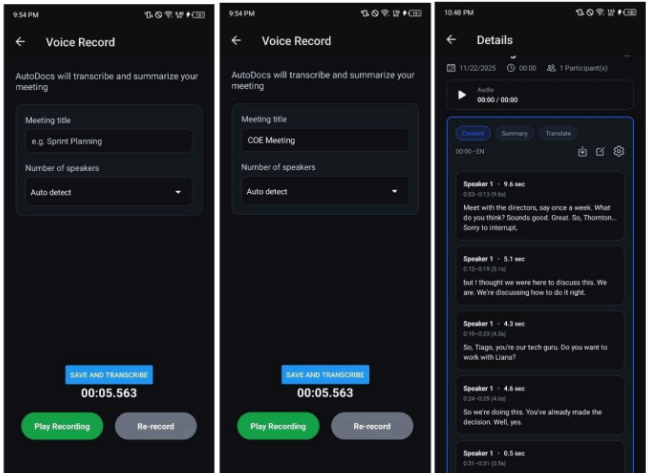


Figure 8. Save and Transcribe

Import existing audio files

Importing existing audio files brings those content already recorded elsewhere into the system for a shared, centralized locus of all relevant audio material. It takes care of such talks, presentations, or interviews that existed beyond the pale of the platform but need to be saved and organized efficiently. It supports various audio formats and lends itself to other sources and devices so that it can rope users into covered materials or historical recording within one repository that they can access without fuss.

It also enables users to produce workflow continuity, reference past recordings while adding new content, and be able to distribute the work better in a team. It promotes flexibility in bringing team members together again and again into an important discussion or presentation without having to source the original recording they were looking for separately. It addresses the close scrutiny of an organized streamlined method of managing the audio resources within an organization (Reda & Chidiac, 2025).

In Figure 9, Import Audio Interface, on the homepage, the feature "Add" appears, and clicking it includes "Import" in the pop-up menu leading to a page wherein the user can upload a certain audio file from his device while adding recorded files into the system to be used later and linked for management.

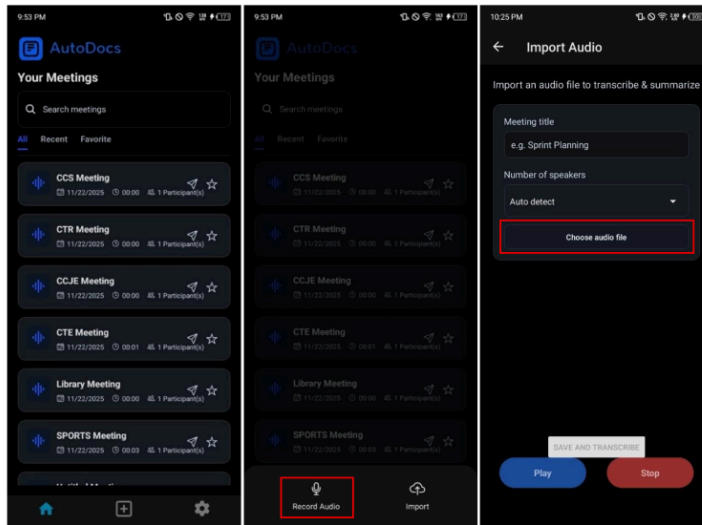


Figure 9. Import Audio Interface

The audio file can be selected, and users are then shown the file storage section within the system where the imported file is present. At this moment, the users may fill in any necessary data fields related to the audio file to record all relevant information. After going through the necessary entries, the audio file can be saved, becoming safely stored within the platform for easy access, management, and reference in the future.

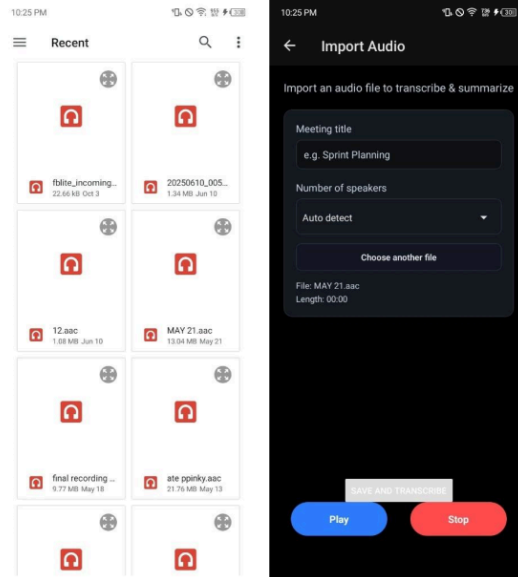


Figure 10. Select and Save Audio

To integrate speech-to-text transcription with number of speaker identification

In figure 11, Speech-to-Text with Speaker Identification, demonstrates speech transcription capabilities together with identifying several speakers in a single audio recording. It enables the application to convert any spoken text into written words, alongside identifying different speakers as the said statement is attributed to a particular individual. This provides clarity and context in the generated transcripts when following conversations, meetings, or presentations of multiple participants. Pragmatic value is manifested, particularly in professional or collaborative environments that necessitate documentation and accountability of the "who said what" (Wang et al., 2024).

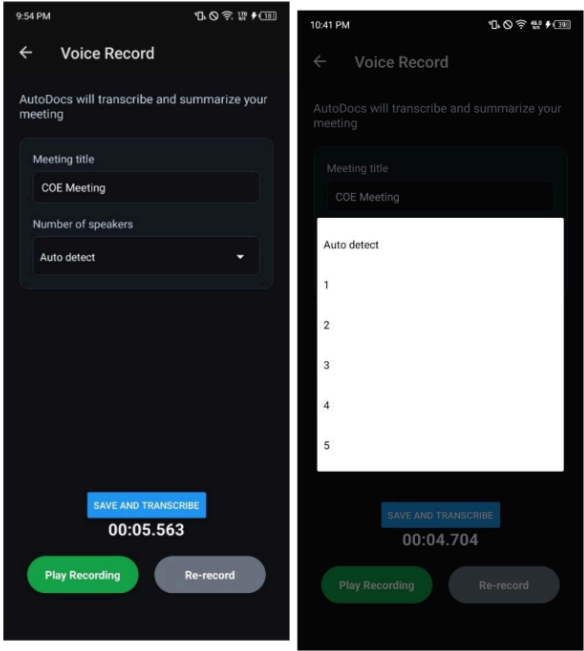


Figure 11. Speech-to-Text with Speaker Identification

To implement features that enable users to edit transcripts for accuracy and clarity, summarize content into concise reports, and translate documents between English and Filipino

Summarization abstracts meeting contents into brink-action reports making it useful because in advocacy where time is precious, turns very clear to the stakeholders to process action. Also, the language translation makes it more accessible because it includes both English and Filipino languages, which makes the documentation really fit for a multilingual realm, and ensures inclusivity and comprehensibility to its users. All of this has diversified the inclusion of raw audio as recorded, concise active documents easy for communication and efficient decision-making.

Edit transcripts for accuracy and clarity

In the feature Edit transcripts to accuracy and clarity of Fig. 12, users can fine-tune the already generated text so that it reflects the content of the audio accurately. Although transcription using AI is a fast and convenient means of converting speech into text, it can yield incorrect results due to accents, background noise, and overlapping speech. Users will now be able to check and edit those inaccuracies. This means that the accuracy, reliability, and the whole meaning of the final transcript will be preserved and irrevocably rendered understandable.

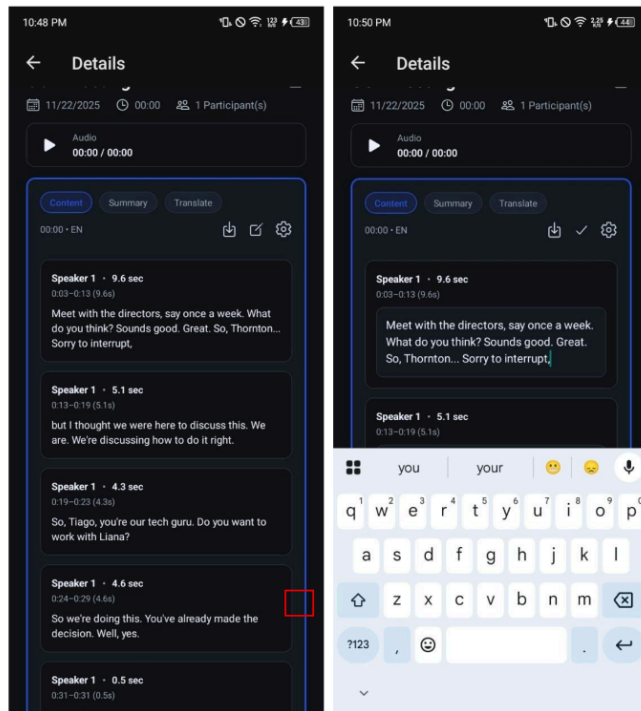


Figure 12. Editing Transcription

Summarize content into concise reports

The summarized report also increases clarity and accessibility of these essential insights to all intended audiences. These concise summaries can easily be shared with team members, management, and external partners so that they are making well-informed decisions and collaboratively working together. Furthermore, this guarantees a well-organized documentation process, providing intelligible and easily accessible records of discussions for future reference. Therefore, the generation of short reports

facilitates the transformation of detailed meeting content into action items and intelligible documentation (Mamdouh et al., 2025).

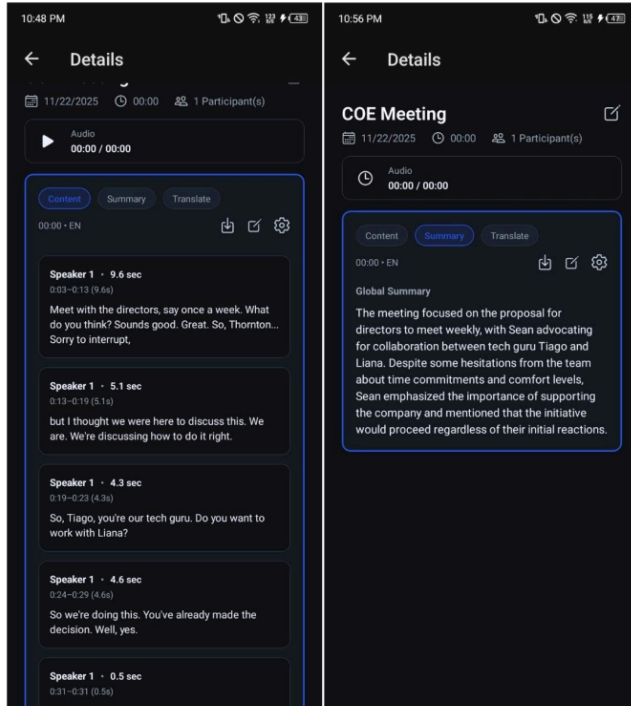


Figure 13. Summarized Reports Interface

Translate documents supporting both English and Filipino languages

An interface for document translation (Figure 14) shows that this system can translate documents in both English and Filipino. The system provides users with a straightforward method to translate text between languages which enables them to share important information with multiple audiences. The platform's translation service eliminates the requirement for external translation tools which enables users to work more efficiently while maintaining uniformity in their documentation process. The tool

benefits organizations and academic institutions by enabling transparent communication of objectives through language translation (Ataman et al., 2025).

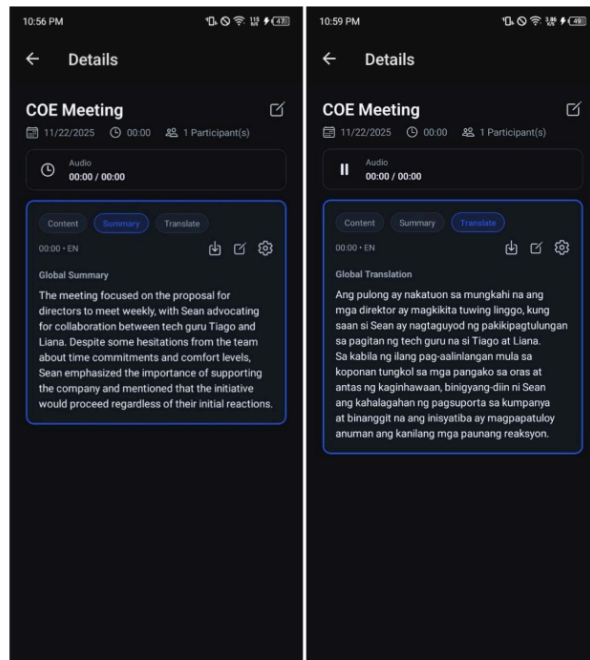


Figure 14. Document Translation Interface

To provide functionalities for exporting reports to various file formats and tracking meeting duration

You have acquired your training materials until the month of October in the year 2023. The system Export and Meeting Duration Interface allows users to export reports in multiple file formats while tracking meeting times which provides them with tools for documentation and time management. Users can export and save the reports in an Audio, PDF, Word, or Text format for easy sharing, archiving, or further editing outside of the platform. The organization requires this method to distribute and store meeting

outcomes and summaries which enhances accessibility for multiple work processes (Oliveira & Oliveira, 2025). The system provides users with meeting duration tracking together with export functions which enable users to monitor meeting times and record all time spent during sessions. The tool helps organizations which need to measure productivity and want to schedule follow-up meetings and track their time usage. The platform enables complete documentation and time tracking through its export function together with its duration monitoring feature which ensures that meeting records support decision-making and organizational planning activities (Avornicului et al., 2024).

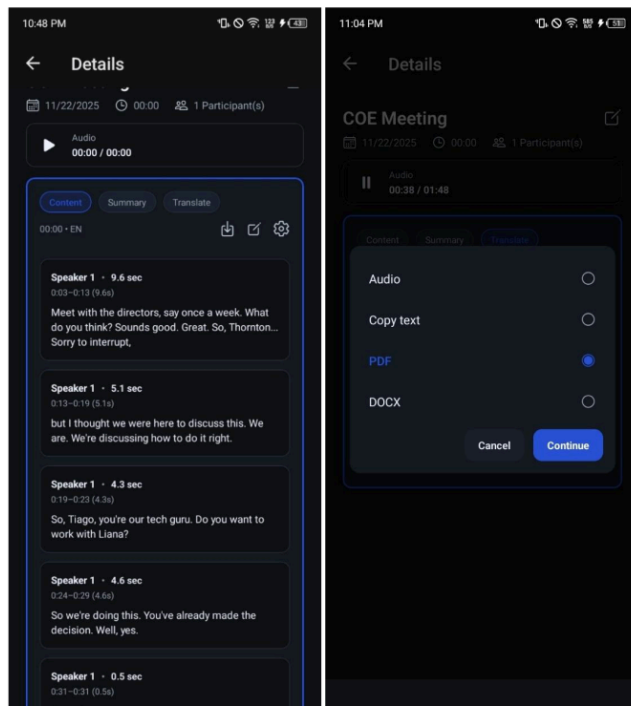


Figure 15. Export and Meeting Duration Interface

To evaluate the developed system using the Technology Acceptance Model (TAM), focusing on perceived ease of use, perceived usefulness, attitudes toward use, and behavioral intention to use

The Technology Acceptance Model (TAM) served as the evaluation framework for the AUTO DOCS system, enabling researchers to assess user reactions to the system and its components. The application presents an intuitive operation experience because users can navigate the system without difficulties, which creates a low barrier for first-time users. The system proved to be useful for documentation tasks according to users, which demonstrated its value for real professional work. Attitude toward use looked at overall satisfaction and willingness to adopt the system from the users' perspective, while behavioral intention to use was an indication of the users' likelihood to continue with the engagement and subsequently integrate the system into their everyday workflows. Such metrics based on TAM gave a holistic view on acceptance of any tool into working environment, measures of its effectiveness and the extent of positive influence on productivity of potential users.

Perceived Ease of Use

Table 4. EOU of AutoDocs: In evaluating the accessibility, accuracy, and interaction in the transcription and documentation features of the system, users deemed AutoDocs to be extremely user-friendly, with an average weighted mean of 4.56 under the verbal interpretation category of "Highly Acceptable". In particular, users rated high their appreciation for being able to access transcription results and information without difficulty, having the correctness of transcription data ensured, and gaining accurate insights. From these findings, it can be inferred that AutoDocs is, in fact, aligning with the Technology Acceptance Model, where perceived ease of use is regarded as one of the most essential attributes before technology adoption, and the expectations of users

with regard to easy navigation and trustworthy performance are being met (Umar et al., 2025).

An item analysis, on the one hand, shows micro-differences among users on such issues as error and inconsistency avoidance in the transcription details. On the other hand, some items scored lower than others. However, there seems to be a level of overall agreement that the system helps ease the cognitive load on users when carrying out documentation tasks, thus enhancing productivity and confidence in output. The study results further verify that AutoDocs makes interaction with meeting and presentation content easy while enhancing user satisfaction, thus increasing its likelihood of acceptance in business and educational settings (Liu et al., 2022).

Table 4. Perceived Ease of Use of AutoDocs

PERCEIVED EASE OF USE	WEIGHTED MEAN	VERBAL INTERPRETATION	RANK
PEOU1: Using AutoDocs allows me to access transcription results and information without difficulty.	4.8	Highly Acceptable	1
PEOU2: Using AutoDocs ensures that all transcription data are correct.	4.6	Highly Acceptable	3
PEOU3: Using AutoDocs minimizes errors and inconsistencies in transcription details.	4.2	Acceptable	5
PEOU4: Using AutoDocs provides accurate insights.	4.6	Highly Acceptable	3
PEOU5: Using AutoDocs ensures that the information reflects the real transcriptions.	4.6	Highly Acceptable	3
Average Weighted Mean	4.56	Highly Acceptable	-

Perceived Usefulness

Table 5 shows the perceived usefulness of AutoDocs by participants in terms of productivity, task efficiency, and contribution to overall meeting management. The system received a 4.73 average rating from users which shows that users consider the

system to be highly acceptable. Users have particularly observed that AutoDocs enables them to complete meeting tasks with greater ease while their meeting management abilities and productivity in planning meetings have both improved (Wali et al., 2023). The system operates as an effective tool for secretarial work because it assists with documentation tasks which enables users to perform their meeting management responsibilities.

The evaluation also mentioned that AutoDocs helps with transcription accuracy and utility in managing meeting content, giving further credence to the system in terms of its practical value in real-world contexts. The evaluation showed that different user groups experienced benefits from the system at different levels which resulted in slight differences between their item rating decisions. The appraisal was, therefore, unanimously positive which indicates that AutoDocs improves workflow management while multiple users gain work-related benefits from it (Hou & Sing, 2025).

Table 5. Perceived Usefulness of AutoDocs

PERCEIVED USEFULNESS	WEIGHTED MEAN	VERBAL INTERPRETATION	RANK
PU1: Using AutoDocs helps me accomplish meeting tasks more efficiently.	4.9	Highly Acceptable	2.5
PU2: Using AutoDocs improves my performance in handling organizational meetings.	4.9	Highly Acceptable	2.5
PU3: Using AutoDocs increases my productivity in organizing meetings.	4.9	Highly Acceptable	2.5
PU4: Using AutoDocs contributes to better transcription within all meetings.	4.6	Highly Acceptable	5
PU5: Using AutoDocs makes it easier to manage meetings.	4.9	Highly Acceptable	2.5
Average Weighted Mean	4.84	Highly Acceptable	—

Attitudes Toward Use

The data shown in Table 6, Attitudes Toward Use, shows how respondents understand AutoDocs according to its system design and convenience and time-saving features. The results show that most respondents found the system easy to navigate because they gave accessibility statements a weighted mean score of 4.9 and they found all tasks performed through the system to be simple and they experienced overall system convenience. The 4.6 rating shows how AutoDocs helps users through its instructions and system feedback which leads to better user experiences. The system achieves high user acceptance according to its weighted mean score of 4.84 which leads to positive user attitudes about AutoDocs application (Sousa et al., 2024).

The data shows that respondents considered time-saving benefits of AutoDocs to be valuable because they could complete meeting documentation work with maximum efficiency. High scoring for all statements indicates a strong agreement that the system alleviates cognitive load from users who need to help secretaries with transcribing and summarizing meetings. The research results demonstrate that AutoDocs operates according to the TAM model which creates positive user attitudes that drive system adoption and ongoing user interaction with the platform (Liao et al., 2022).

Table 6. Attitudes Toward Use of AutoDocs

ATTITUDES TOWARD USE	WEIGHTED MEAN	VERBAL INTERPRETATION	RANK
ATU1: Using AutoDocs is easy to understand and navigate.	4.9	Highly Acceptable	2.5
ATU2: Using AutoDocs allows me to perform tasks without difficulty.	4.9	Highly Acceptable	2.5
ATU3: Using AutoDocs provides clear instructions and system feedback.	4.6	Highly Acceptable	5
ATU4: Using AutoDocs feels convenient and user-friendly.	4.9	Highly Acceptable	2.5

ATTITUDES TOWARD USE	WEIGHTED MEAN	VERBAL INTERPRETATION	RANK
ATU5: Using AutoDocs saves me time when performing meeting tasks.	4.9	Highly Acceptable	2.5
Average Weighted Mean	4.84	Highly Acceptable	—

Behavioral Intention to Use

As seen in Table 7, concerning the respondents' behavioral intentions toward the adoption of AutoDocs, the system were viewed as relevant to organizational goals, support for existing workflows, and aid in fulfilling institutional objectives. Accordingly, results demonstrate that participants generally or strongly believed that the system would meet the operational and technical requirements of their offices. Across statements, weighted means were mostly in the range of 4.4 to 4.8, indicating the secretaries viewed the system as assisting them in executing their tasks and meeting the requirements of documentation. The overall average weighted mean of 4.64 also indicates that, in respect of fostering the behavioral intention to use the system among end users, it has an extremely good acceptance (Kusonwattana et al., 2022).

Furthermore, AutoDocs is considered a tool that is effective and strategic for augmenting existing processes while remaining aligned with institutional standards. Respondents believe that the system helps achieve objectives of accurate and efficient meeting summaries and therefore increases their confidence in the adoption of AutoDocs as part of their daily workflow. In sum, AutoDocs fares well in terms of supporting operational efficiency and organizational applicability, two things that are heavily weighed in encouraging sustained use and continued adoption (Durão & Palma dos Reis, 2025).

Table 7. Behavioral Intention to Use

BEHAVIORAL INTENTION TO USE	WEIGHTED MEAN	VERBAL INTERPRETATION	RANK
BIU1: Using AutoDocs aligns with the goals and needs of all secretaries.	4.6	Highly Acceptable	4
BIU2: Using AutoDocs supports existing workflows and processes at LSPU.	4.6	Highly Acceptable	4
BIU3: Using AutoDocs helps fulfill the objectives of the institution's meeting summary.	4.8	Highly Acceptable	1
BIU4: Using AutoDocs aligns with the standards and guidelines of the university.	4.5	Highly Acceptable	5
BIU5: Using AutoDocs fits well with the organization's operational and technological needs.	4.7	Highly Acceptable	2
Average Weighted Mean	4.64	Highly Acceptable	—

Overall Evaluation of AutoDocs Based on TAM Criteria

Table 8 outlines the general evaluation of AutoDocs based on the criteria of the Technology Acceptance Model (TAM). Results show that users accepted the system very much for all criteria, while the highest weighted means were found in the categories of Perceived Usefulness (4.84) and Attitudes Toward Use (4.84), which suggest that the secretaries from LSPU Siniloan Campus appreciated both the system's applicable benefit and user-friendliness with its efficient capability in speeding up the transcription and documentation processes.

The users found the system to be intuitive because they could use the interface through basic navigation between different screen locations. The design and usability features of AutoDocs enabled users to work more comfortably while they learned to use the system through clear instructions and real-time transcription and editable summary functionality. The perceived usefulness rating of 4.84 shows that users recognized the actual economic advantages which came from improved productivity and decreased errors and easier meeting documentation work (Lekova et al., 2025).

The system received high user approval because it achieved 4.72 as its overall grand mean score which demonstrated strong system performance. The development approach receives validation through the high TAM criterion ratings which especially show the effectiveness of OpenAI Whisper and large language models for transcription and summarization. AutoDocs functions as a tool which enhances the documentation processes of meetings in educational and business environments because it helps organizations work more efficiently while they collect essential facts.

Table 8. Overall Evaluation of AutoDocs Based on TAM Criteria

CRITERIA	WEIGHTED MEAN	VERBAL INTERPRETATION	RANK
Perceived Ease of Use (PEU)	4.56	Highly Acceptable	4
Perceived Usefulness (PU)	4.84	Highly Acceptable	1.5
Attitudes Toward Use (ATU)	4.84	Highly Acceptable	1.5
Behavioral Intention to Use (BIU)	4.64	Highly Acceptable	3
Grand Mean	4.72	Highly Acceptable	—

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATION

Summary

The system function now transforms spoken language into written text through its improvement which integrates speech-to-text technology and multiparty speaker identification capabilities for precise speech-to-text conversion because each speaker's voice gets correctly assigned to their respective identity. This system creates separate sections for user input which improves transcription accuracy by providing specific details and background information. The study demonstrates that this function enhances their ability to organize information while improving their efficiency for locating specific elements and connecting duties to particular parts of text. AUTO DOCS creates a better system which avoids mistakes and improves responsibility tracking through its speaker-based text organization system that produces complete and precise documents which users can take action on.

The system maintained its high usability and effectiveness because of its three core functionalities which include transcription editing and content summarization and document translation. The transcription process enables users to fix mistakes and change sentence structures while making content easier to read which results in accurate transcriptions of the original audio content. The process of summarization transforms lengthy conversations into brief reports that contain essential information which users can quickly review to find important topics and resolutions and tasks that need to be completed. The translation system allows users to communicate effectively in multilingual environments by supporting both English and Filipino while making it

easier for others to understand. Ameanor: The features enable users to produce professional documents from ordinary audio recordings which improve document functionality because of their better operational performance and enhanced text understanding.

The system provides users with dual functions which permit them to send reports through different file formats while users can also use the system to monitor their meetings through its meeting duration tracking feature. Users can store reports in audio, pdf, word, and text format because they can share their reports with others while preserving their ability to store their reports and transform their reports into different formats. The meeting duration tracking system ensures precise record keeping of time spent on each session because it records every session's actual time. The system enables organizations to record their meetings and create sound recordings while maintaining their audio quality throughout all of their meeting operations.

AUTO DOCS received high user acceptance from its assessment through the Technology Acceptance Model (TAM) system evaluation. The system received its highest ratings for Perceived Usefulness (4.84) and Attitudes Toward Use (4.84) which measured the material advantages of the system plus its user-friendly design while users showed readiness to adopt the system according to their Behavioral Intention to Use (4.64) and Perceived Ease of Use (4.56) results. The average score of 4.72 demonstrates how AUTO DOCS delivers its expected performance to users through its direct application of OpenAI Whisper and advanced AI systems which transform data into transcription and summary and documentation work.

Conclusion

Thus, the live capture and importation of existing sound files into AUTO DOCS is a significant step in the changing practices in documentation. Speech content would be effectively captured and classified in such a manner that only the pertinent information would be relatively easy to find, retrievable, and secured. The very users would enjoy a simplified work process that would save on errors, would be supportive of accurate record keeping, and would provide constant follow-up on key points, action items, and decisions. Its ability to handle real-time and previously recorded audio reflects great flexibility in adapting to professional and organizational diversity.

Further, the functionality of report outputting to varied file formats such as Audio, PDF, Word, or Text is provided for exporting the documents produced within the system. Infinity enables the user to share, save permanently, or undergo post-editing of a transcript. Tracking meeting duration gives precise data on how much time each session lasts, so the companies can use this information to check productivity levels and effectively allocate resources. All in all, these capabilities contribute to improved organization, effective, and most practical documentation and time management processes.

Overall perception of users confirms that AUTO DOCS is highly valued and accepted. The high perceived usefulness, attitudes toward use, and good behavioral intention to use and ease of use indicate how well the system satisfies users' expectations and improves adoption in the regular workflow. In total, AUTO DOCS stands out as a platform dependable, user-friendly, and highly effective, where productivity, accuracy in documentation, and organization in meeting documentation

are respected. Which makes the system a worthwhile resource in professional and educational settings.

Recommendation

To ensure the application of all features of remote recording and recording import for adequate data, continuous usage toward organized storage and assigning exactly processed fields in the presence of recordings, is a mandate. Inconsistent performance from either the users or the organization will contribute to improper use and storage of audio data in the platform.

Training users in conjunction, because of the transcription with laced speaker identifiers, will further ease the workflow process, favoring the informed decision-making and recognition of the value of a group member's input and, needless to say, some constructive action for their inputs.

For users to improve the quality, clarity, and accessibility of documentation, it is suggested that utilization of editing, summarization, and translating features be made compulsory during interaction with the system. Practices for post-editing transcripts, writing concise reports, and translating materials as needed, especially within any given corporate standard, should be established. Training of chosen features will quickly build skills in users, greatly contribute to punctual documentation, and allow communication of more information within teams and wherever communication is multilingual.

Large utilization of the results of exporting, and during the meetings, tracking the duration will be a sure help in enhancing their experiences with the app. The organizations are thereby supported to begin archiving reports in preferred formats and

reviewing a certain amount of meeting duration. The knowledge of these functionalities will aid in the solid exploitation of the system for cogent documentation, reporting, and organization planning.

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